

LU ZHUOYI

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EDUCATION

- Aug 2022 – Jun 2026 **Nanyang Technological University, Singapore** (CGPA: 4.73/5.00, Top 5%)
Bachelor of Mechanical Engineering
Robotics and Mechatronics Stream, Minor in Mathematics
- Jun 2025 – Aug 2025 **Rice University, Houston, Texas, United States of America**
Summer Research Intern @RobotPI Lab under Prof. Kaiyu Hang
- Sep 2016 – Jun 2022 **Desheng School, Guangdong, China**
International Baccalaureate Diploma (IBDP): 41/45

SKILLS

- Programming & Tools** Python, Anaconda/Conda, MATLAB, C/C++
- Robotics & Middleware** Linux OS, Neutrino OS, ROS1/2, MuJoCo, Maniskill
- Hardware & Embedded** Arduino, STM32, SolidWorks, Autodesk Fusion

RESEARCH EXPERIENCE

- Robotics Research Centre (RRC), NTU Singapore** May 2024 – Present
- Research on Series-Elastic Actuator model of HEBI motor. Designed experiments for motor parameter estimation e.g., Armature (Torque-Current), Velocity Constant (Voltage-Velocity) with PWM and PID control.
 - Built dual planar system consisting of two 2-DoF robots and 6-DoF arms with HEBI motors. Establish Teleoperation based Impedance control by deriving its kinematics and dynamics.
 - Research on Motion Planning based on Manipulation Potential under Quasi-Static Assumption. Trajectory/Dynamic Movement Primitives (DMP) Optimization based on Black-Box Optimization (BBO).
- Rice University, Houston, Texas, United States of America** Jun 2025 – Aug 2025
- Develop the codebase (e.g., Keyboard, XBOX joystick and VR Teleoperation Control) for XLeRobot: Fully Autonomous Household Dual-Arm Mobile Robot for \$660 (<https://github.com/Vector-Wangel/XLeRobot>).
 - Autonomous Grasping with RGB/Segmentation inputs trained by Reinforcement Learning (PPO) via ManiSkill and zero-shot real-world deployment. Developed based on Stones's LeRobot Sim2Real codebase.
- Delta Electronics, Singapore** Jan 2025 – May 2025
- Modeling quadruped jumping using virtual mass-spring-damper attached between each leg's hip and foot frame. Validate in both simulation (Gazebo and Mujoco) and real-world robot Unitree Go2.
 - Perform real-world navigation between nodes on a 2D map using ROS2 Nav2 on a mobile robot.
 - Integrate Mujoco with Pinocchio Library as a basic kinematics solver for Manipulation tasks in C++.

PEER-REVIEWED PUBLICATIONS

- Yang, L., Turlapati, S. H., **Lu, Z.**, Lv, C., & Campolo, D., “A Planning Framework for Stable Robust Multi-Contact Manipulation.” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Hangzhou, China, Oct. 2025.

MANUSCRIPT

- **Lu, Z.**, Yang, L., Turlapati, S. H., Campolo, D., “Haptic Sorter: A Unified Planning Framework for Online Shape Estimation and Real-Time Pose Inference.” arXiv preprint arXiv:2605.31352.

PROJECTS

MA4830 Realtime Software For Mechatronic Systems Jan 2026 – May 2026

- Leader and Main Programmer for all coding projects in the Neutrino OS in C language.
- Leveraging RTOS techniques (i.e., Multi-Threads Communication and Synchronization) for Waveform Generation output from a DAS16 board to an oscilloscope, with an UI and high control frequency of $100Hz$.

Undergraduate Research Program (URECA) under Prof. K Jimmy Hsia Jul 2023 – Jun 2024

- Developed a new frontal photopolymerization strategy for functionally-graded materials based on UV attenuation. Published in URECA Papers and Posters.

Kaggle Competition in Modeling earthquake richter and impacts Aug 2023 – Dec 2023

- Model earthquake damage grades in Nepal using location and construction data from Kaggle Dataset.
- Data cleaning, one-hot encoding, visualization, Random Forest, Logistic Regression, and KNN applied.

EXTRACURRICULAR ACTIVITIES

Xinyuan Charity Center, Guangdong, China Aug 2020 – Aug 2021

- Organized drawing lessons for autistic children to help them adapt to school and build social relationships.

Residential Hall Garden Leader, NTU Singapore Aug 2022 – June 2026

- Led weekly gardening activities, including seed germination, watering, and tool maintenance.

HONORS AND AWARDS

- Dean's List Certificate AY2023/24 (Top 5%), Mechanical Engineering, NTU Singapore.
- Dean's List Certificate AY2024/25 (Top 5%), Mechanical Engineering, NTU Singapore.
- URECA Undergraduate Research Program 2023/2024 Certificate.
- Gold Medalist (Top 8%) in the British Physics Olympiad (BPhO), 2021.